

Notes: Bénabou and Tirole (QJE, 2002)

Self-Confidence and Personal Motivation

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
2.1	Contribution relative to standard economics	2
2.2	Contribution relative to time-inconsistency models	2
2.3	Contribution relative to psychology	3
2.4	Contribution relative to models of beliefs as utility	3
2.5	Contribution to organizational economics	3
2.6	Contribution to behavioral economics	3
3	Model objects and measurement	3
3.1	Ability, effort, and the task decision	3
3.2	Quasi-hyperbolic preferences and conflict between selves	3
3.3	Information structures	4
3.4	Self-handicapping	4
3.5	Endogenous memory	4
4	Models and theoretical specifications	4
4.1	Value of information: confidence maintenance versus overconfidence	4
4.2	Proposition 1: Who wants to avoid information?	5
4.3	Self-handicapping condition	5
4.4	Memory equilibrium and the recall rate	5
4.5	Proposition 2: Multiple equilibria in memory management	5
4.6	Proposition 3: Costly memory and awareness management	6
4.7	Proposition 4: Welfare of self-deception	6
4.8	Defensive pessimism	6
5	Identification and theoretical design logic	6
6	Section-by-section study guide	7
6.1	Introduction	7
6.2	Section I: The Demand for Self-Confidence	7
6.3	Section II: Self-Deception and Endogenous Memory	7
6.4	Section III: Beliefs and Make-Beliefs	7
6.5	Section IV: Welfare Analysis of Self-Deception	7
6.6	Section V: Variants and Extensions	7

6.7	Conclusion	7
7	Figures and theoretical result blocks	8
7.1	Figure I: Memory equilibria as a function of present bias	8
7.2	Figures IIa and IIb: Costly memory or awareness management	8
7.3	Figure IIc: Low- β case	9
7.4	Theoretical result block 1: Demand for confidence	10
7.5	Theoretical result block 2: Information can be harmful	10
7.6	Theoretical result block 3: Endogenous memory	10
7.7	Theoretical result block 4: Welfare	11
7.8	Theoretical result block 5: Defensive pessimism	11
8	Detailed result map	11
9	Short summary	11
10	Conclusion	12
10.1	Core conclusion	12
10.2	Economic intuition	12
10.3	Incremental contribution revisited	12
10.4	Implications	12
10.5	Limitations	12
10.6	Takeaway	12

1 Big picture

- **Question.** Why do rational people often value self-confidence, avoid information about themselves, selectively remember successes, self-handicap, or maintain overly positive beliefs even when accurate beliefs would seem decision-theoretically useful?
- **Core economic idea.** Self-confidence can be instrumentally valuable because ability and effort are often complements. A person who believes success is likely is more willing to exert costly effort or persevere in a project.
- **Time-inconsistency channel.** The paper models a conflict between a date-0 self and a date-1 self. Date-1 has excessive present bias and may underinvest effort. Date-0 may therefore want date-1 to hold optimistic beliefs, because optimism offsets procrastination or weak willpower.
- **Demand for self-confidence.** People may value self-confidence for three reasons: hedonic consumption value, signaling value to others, and motivational value. The paper emphasizes the motivation channel because it generates endogenous and state-dependent demand for self-serving beliefs.
- **Supply of self-confidence.** The paper models endogenous memory and awareness management. A person can strategically encode, repress, or recall information in ways that shape future beliefs.
- **Bayesian self-deception.** The key innovation is that self-deception need not require irrational inference. The future self is Bayesian and understands that memory may be selective; this metacognition creates an intrapersonal signaling game.
- **Multiple equilibria.** Selective memory creates strategic complementarity between date-0 repression and date-1 skepticism. This often generates multiple personal equilibria: truth-telling, partial repression, and full repression.
- **Welfare ambiguity.** Positive thinking can improve welfare by overcoming self-control problems, but it can also be self-defeating by causing overconfidence, nonproductive persistence, or costly memory manipulation.
- **Broader lesson.** The paper brings psychology into economics by showing how motivated beliefs, self-handicapping, defensive pessimism, and self-serving memory can arise from rational intertemporal incentives.

2 Incremental contribution of the paper

2.1 Contribution relative to standard economics

Standard decision theory treats information as weakly valuable and assumes that beliefs are disciplined by Bayesian updating. Bénabou and Tirole show that information can be instrumentally harmful when future selves have imperfect willpower. Accurate information may reduce the future self's motivation exactly when the date-0 self wants the future self to persist.

2.2 Contribution relative to time-inconsistency models

Carrillo and Mariotti-style strategic ignorance shows that time-inconsistent agents may avoid information. This paper broadens the logic from ignorance to self-confidence, selective memory, self-handicapping, and intrapersonal communication. The result is not merely “do not learn.” It is a general theory of how the self manages beliefs over time.

2.3 Contribution relative to psychology

Psychology had long documented overconfidence, selective recall, and self-esteem maintenance. The paper contributes a formal economic model that organizes these facts using incentives, beliefs, and strategic communication. It shows how apparently irrational phenomena can be consistent with Bayesian reasoning once the self is modeled as multiple temporally separated decision makers.

2.4 Contribution relative to models of beliefs as utility

Akerlof-Dickens and related models often put beliefs directly into utility. Bénabou and Tirole provide an endogenous instrumental value of beliefs. Self-confidence is valuable because it changes future effort choices. Hedonic self-image can be added, but it is not necessary for self-deception to emerge.

2.5 Contribution to organizational economics

The paper shows why teachers, parents, managers, coaches, partners, or coworkers may invest in others' morale. If self-confidence affects effort, agents who benefit from someone's success have incentives to manipulate that person's self-perception.

2.6 Contribution to behavioral economics

The paper offers a unified economic framework for several behavioral phenomena: optimism, strategic ignorance, self-handicapping, memory repression, positive illusions, defensive pessimism, and personal rules. It is one of the canonical papers showing how belief distortions can be endogenously generated rather than simply assumed.

3 Model objects and measurement

3.1 Ability, effort, and the task decision

An individual has uncertain ability θ , interpreted as the probability that a project succeeds if the individual exerts effort. At date 1 the individual chooses whether to exert costly effort $c > 0$. Success yields benefit V at date 2; failure yields no benefit. Self-confidence is summarized by the expected value of ability under the individual's belief distribution:

$$\theta_1 = \int \theta dF_1(\theta).$$

Interpretation: Ability and effort are complements in the baseline model: when perceived ability is higher, the expected return to effort is higher. This is the functional value of self-confidence.

3.2 Quasi-hyperbolic preferences and conflict between selves

The date-1 self undertakes the activity when

$$\theta_1 > \frac{c}{\beta\delta V},$$

while the date-0 self wants the date-1 self to undertake it whenever

$$\theta_1 > \frac{c}{\delta V}.$$

Because $\beta < 1$, the date-1 self has a higher effort threshold and is too prone to quit or procrastinate.

Interpretation: The time-inconsistency wedge creates the instrumental demand for optimism. Confidence can make date-1 undertake projects that date-0 values but date-1 would otherwise abandon.

3.3 Information structures

At date 0 the individual can choose whether date-1 will learn ability precisely, receive noisy feedback, or remain uninformed. Information has two opposing effects:

- it reduces overconfidence and prevents bad projects from being pursued;
- it may destroy motivation by lowering confidence into the time-inconsistency region.

Interpretation: In classical decision theory, the first effect dominates and information is valuable. With present bias, the second effect can dominate, and information can be avoided.

3.4 Self-handicapping

Self-handicapping is modeled as choosing an inefficient or uninformative action at date 0. It destroys information about ability and can therefore preserve future motivation. Examples include choosing excuses, deliberately reducing preparation, taking overly difficult tasks, or avoiding diagnostic feedback.

Interpretation: Self-handicapping is privately costly but can serve as a commitment device: it protects future self-confidence from bad news.

3.5 Endogenous memory

The supply side of self-confidence is modeled as a strategic memory/awareness system. Date-0 can influence the probability that negative information is encoded or later recalled. Let λ denote the probability that adverse information is remembered or transmitted to the future self. A smaller λ means greater repression.

Interpretation: Memory becomes a communication channel between selves. Date-0 is the sender; date-1 is the receiver. Because the receiver is Bayesian, the absence of bad memories is itself informative.

4 Models and theoretical specifications

4.1 Value of information: confidence maintenance versus overconfidence

If the date-1 self would undertake the task absent information, the value of fully learning ability can be decomposed as

$$I_F = G_F - L_F,$$

where G_F is the gain from avoiding overconfidence and L_F is the loss from falling into the time-inconsistency region.

Interpretation: G_F pushes toward information; L_F pushes toward ignorance. Present bias raises the importance of L_F .

4.2 Proposition 1: Who wants to avoid information?

Proposition 1: confidence maintenance

If an individual prefers not to receive information, then another individual with a distribution of ability that dominates his in the monotone likelihood ratio sense also prefers not to receive information. In words: sufficiently confident people have more to lose from unfavorable information, so they may be more willing to protect self-confidence.

Interpretation: This formalizes the “more to lose” effect. People with high priors may defend their ego because bad news threatens a valuable motivational asset.

4.3 Self-handicapping condition

Self-handicapping occurs when the motivational benefit of avoiding diagnostic information exceeds the direct cost of the handicapping action:

$$\text{motivational value of ignorance} > \text{cost of self-handicapping.}$$

Interpretation: The model explains why self-handicapping is more likely when willpower problems are strong and handicapping costs are small. It also explains why both high-confidence and low-confidence people may self-handicap for different reasons.

4.4 Memory equilibrium and the recall rate

Let λ^* denote the equilibrium recall/transmission probability for bad news. Date-1 discounts the absence of bad news if he knows that date-0 may have repressed it. This creates a self-referential equilibrium condition:

$$\lambda^* \in \{0, \Lambda(\beta), 1\}$$

for intermediate values of present bias.

Interpretation: Full repression can be self-confirming because absence of bad news is expected. Full truthfulness can also be self-confirming because absence of bad news is credible. Partial repression occurs between these extremes.

4.5 Proposition 2: Multiple equilibria in memory management

Proposition 2: memory equilibria

When memory costs are absent, there exist regions of present bias in which the intrapersonal memory game has multiple equilibria: full repression, partial repression, and full recall. The selection depends on the strength of time inconsistency and the risk of overconfidence.

Interpretation: The multiplicity comes from metacognition. If future self expects censorship, he distrusts silence, which changes date-0's incentive to censor.

4.6 Proposition 3: Costly memory and awareness management

Proposition 3: awareness costs

With costly repression or recall, a higher degree of time inconsistency or a lower cost of repression expands the range of parameters under which memory manipulation and partial repression equilibria exist.

Interpretation: Small changes in psychic costs of memory management can generate discontinuous changes in beliefs and behavior.

4.7 Proposition 4: Welfare of self-deception

Proposition 4: welfare ambiguity

Positive thinking can dominate honesty when the motivational value of self-confidence is high, but it can be worse when overconfidence and the costs of memory manipulation are large. The welfare ranking depends on the balance between self-discipline gains and decision errors.

Interpretation: The model does not say optimism is always good. It explains when self-deception is beneficial, harmful, or pursued even when harmful.

4.8 Defensive pessimism

When ability and effort are substitutes rather than complements, high perceived ability can reduce effort. In this case individuals may prefer low expectations:

$$\frac{\partial^2 \text{payoff}}{\partial \theta \partial e} < 0.$$

Interpretation: This accounts for defensive pessimism: lowering expectations can increase preparation when confidence would otherwise cause coasting.

5 Identification and theoretical design logic

This is a theory paper, so “identification” is analytical rather than empirical. The paper isolates mechanisms by changing one assumption at a time.

- **Demand side:** compare cases with and without time inconsistency, ability-effort complementarity, and hedonic beliefs.
- **Supply side:** compare exogenous information, strategic ignorance, and endogenous memory.
- **Rationality discipline:** future selves are Bayesian and understand the possibility of selective memory.
- **Welfare analysis:** compare equilibria with repression, partial recall, and truthfulness.
- **Extensions:** analyze substitutes between ability and effort, defensive pessimism, and self-image as a consumption good.

Interpretation: The paper’s theoretical design is powerful because it shows that self-deception can emerge without assuming irrational updating or direct utility from false beliefs.

6 Section-by-section study guide

6.1 Introduction

The introduction frames the puzzle: why do people prefer rosy self-views when overconfidence is costly, and how can rational people deceive themselves? It previews the demand for confidence and the supply of self-deception through memory.

6.2 Section I: The Demand for Self-Confidence

This section explains three reasons for valuing self-confidence: consumption, signaling, and motivation. It then models present bias and shows when information is valuable or harmful.

- Key result: information can lower welfare when it undermines motivation.
- Key phenomenon: self-handicapping protects future self-confidence.

6.3 Section II: Self-Deception and Endogenous Memory

This section introduces selective memory. Bad news may be repressed, but the future self knows repression is possible. This creates an intrapersonal signaling game with multiple equilibria.

- Key result: silence can be either reassuring or suspicious depending on equilibrium.
- Key figure: Figure I shows memory equilibria as a function of present bias.

6.4 Section III: Beliefs and Make-Beliefs

This section interprets the model relative to psychological evidence on overconfidence and self-serving beliefs. It shows that biased self-perception can be rationally generated by selective memory.

6.5 Section IV: Welfare Analysis of Self-Deception

This section compares positive thinking and honesty. It shows that self-deception can improve welfare, but also can be self-defeating and nonetheless pursued.

6.6 Section V: Variants and Extensions

The paper studies defensive pessimism when ability and effort are substitutes, and hedonic self-esteem when beliefs directly enter utility.

6.7 Conclusion

The conclusion emphasizes that self-confidence management matters in principal-agent relationships, education, workplace motivation, personal rules, and models of willpower.

7 Figures and theoretical result blocks

7.1 Figure I: Memory equilibria as a function of present bias

Read:

- The vertical axis is the equilibrium recall/transmission probability λ^* .
- The horizontal axis is β , the present-bias parameter.
- For low β , confidence maintenance dominates and full repression can occur.
- For high β , overconfidence risk dominates and truthfulness is more attractive.
- In the middle, multiple equilibria arise, including partial repression.

Detailed interpretation: Figure I is the central visual representation of the intrapersonal signaling game. When willpower is weak, date-0 has a strong incentive to protect date-1 from discouraging news. When willpower is strong, the value of information dominates. In between, both truthfulness and repression can be self-confirming because future beliefs depend on expectations about the memory rule.

Economic message: Self-confidence management can be discontinuous. Small changes in incentives can move a person from honesty to self-deception.

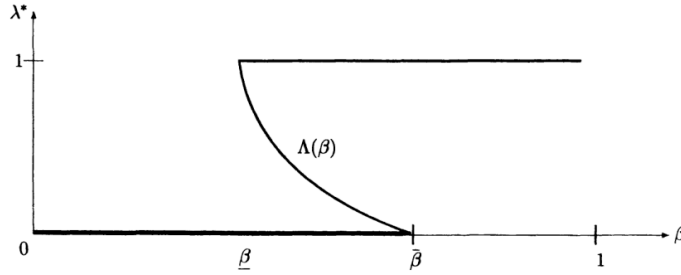


FIGURE I

Figure 1: Figure I: Memory equilibria as a function of present bias.
Single visual: width set to 0.6\linewidth.

7.2 Figures IIa and IIb: Costly memory or awareness management

Read:

- Both figures show equilibrium recall probabilities λ^* as a function of the memory-management cost parameter a .
- Figure IIa corresponds to an intermediate-high range of β .
- Figure IIb corresponds to the most representative intermediate case, where repression, multiplicity, and truthfulness all appear across ranges of a .
- The two boundary functions $\lambda_1(a)$ and $\lambda_2(a)$ define regions with multiple equilibria.

Detailed interpretation: These figures show how small changes in the cost of memory management can create large changes in recall behavior. Low costs can sustain repression. High costs push the agent toward truthfulness. Intermediate costs allow multiple self-fulfilling memory regimes.

Economic message: Awareness management is a technology of self-control. Its cost determines whether people suppress, partially transmit, or honestly preserve information.

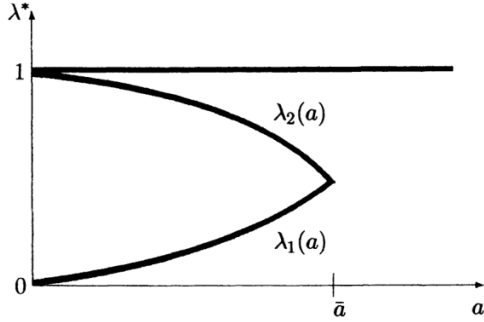


FIGURE IIa
Case $\beta_2 < \beta < \beta_3$

(a) Figure IIa: Case $\beta_2 < \beta < \beta_3$.
Related figure placed on the same row.

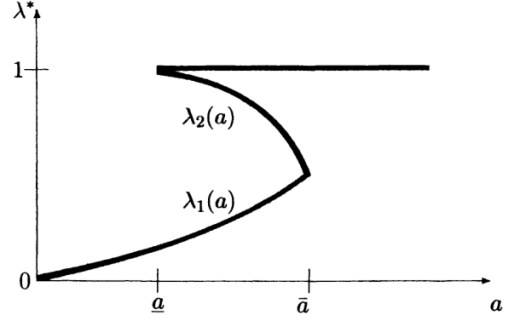


FIGURE IIb
Case $\beta_1 < \beta < \beta_2$

(b) Figure IIb: Case $\beta_1 < \beta < \beta_2$.
Related figure placed on the same row.

7.3 Figure IIc: Low- β case

Read:

- Figure IIc covers the low- β case, where present bias is strong.
- The equilibrium region shifts toward lower recall probabilities.
- The curve $\lambda_1(a)$ increases with memory-management cost.
- For sufficiently high cost a , full recall becomes the equilibrium.

Detailed interpretation: Figure IIc shows the case where self-control problems are severe. In that region, repression has high motivational value and can persist over a wide range of costs. Only sufficiently large costs of repression push the agent toward truthfulness.

Economic message: When willpower is weak, self-deception is more tempting and more robust. The cost of sustaining accurate memory must be high enough to overcome the motivational benefit of optimism.

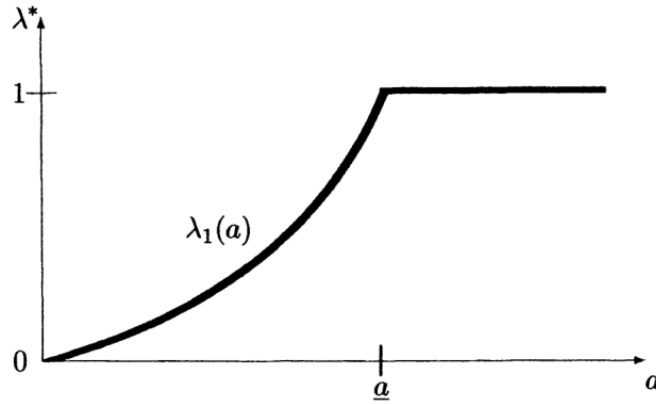


FIGURE IIc
Case $\beta < \beta_1$

Figure 3: Figure IIc: Case $\beta < \beta_1$.
Single visual: width set to $0.6 \backslash \text{linewidth}$.

7.4 Theoretical result block 1: Demand for confidence

Demand for self-confidence

Self-confidence is valuable when it raises future effort. A date-0 self may prefer date-1 to hold optimistic beliefs because date-1 is present-biased and otherwise underinvests effort. This creates a demand for self-serving beliefs without assuming a direct taste for false beliefs.

7.5 Theoretical result block 2: Information can be harmful

Information avoidance

Information has an overconfidence benefit and a confidence-maintenance cost. The overconfidence benefit prevents bad projects; the confidence-maintenance cost is that bad news may push date-1 into procrastination. When the latter dominates, the individual rationally avoids information.

7.6 Theoretical result block 3: Endogenous memory

Selective memory

Self-deception is modeled as a strategic communication game between temporal selves. Date-0 chooses what gets remembered; date-1 interprets absence of bad news knowing that memory may have been selective. This generates multiple Bayesian equilibria.

7.7 Theoretical result block 4: Welfare

Welfare of positive thinking

Positive thinking can be welfare-enhancing when it solves self-control problems, but harmful when it produces overconfidence or costly repression. The theory therefore explains why self-deception can be pursued even when it is ultimately self-defeating.

7.8 Theoretical result block 5: Defensive pessimism

Defensive pessimism

When perceived ability and effort are substitutes, optimism can reduce effort. In that case pessimistic self-beliefs may be useful because they induce preparation and caution. The same framework explains both optimistic self-deception and defensive pessimism.

8 Detailed result map

- **Motivation channel:** self-confidence increases effort when ability and effort are complements.
- **Time-inconsistency channel:** date-0 values optimism because date-1 underinvests effort.
- **Information avoidance:** information can lower welfare by undermining motivation.
- **Self-handicapping:** individuals may choose actions that hide diagnostic information about ability.
- **Memory management:** bad news may be selectively repressed or forgotten.
- **Bayesian self-deception:** future selves rationally interpret memory silence given expected censorship.
- **Multiplicity:** truthfulness, partial repression, and full repression can all be self-confirming.
- **Welfare ambiguity:** optimism can help or hurt; welfare depends on motivation gains versus overconfidence costs.
- **Defensive pessimism:** if confidence reduces effort, pessimism can become motivationally useful.

9 Short summary

- Bénabou and Tirole analyze why rational people may value self-confidence and engage in self-deception.
- Self-confidence is valuable because it motivates effort when willpower is imperfect.
- A present-biased future self may quit too easily, so the current self may prefer the future self to remain optimistic.
- Information can be harmful if it discourages effort more than it improves decision making.
- Self-handicapping protects self-confidence by avoiding diagnostic feedback.
- Selective memory is modeled as a rational intrapersonal communication game.
- Future selves are Bayesian and understand that memory can be strategic.
- Multiple equilibria arise because censorship and skepticism reinforce each other.
- Positive thinking may improve welfare, but may also be self-defeating.
- The framework also explains defensive pessimism when confidence and effort are substitutes.

10 Conclusion

10.1 Core conclusion

Bénabou and Tirole show that self-confidence and self-deception can be understood using economic tools. When people have imperfect willpower, optimistic beliefs can increase motivation. Since beliefs matter for future actions, the current self may rationally manipulate future beliefs through information avoidance, self-handicapping, or selective memory.

10.2 Economic intuition

The key intuition is that accurate beliefs are not always privately optimal when future selves have distorted incentives. A current self may want a future self to work hard, persist, or avoid temptation. If discouraging information would cause the future self to quit too easily, the current self may prefer ignorance or optimism.

10.3 Incremental contribution revisited

The paper's incremental contribution is to endogenize the demand and supply of motivated beliefs. It derives the value of self-confidence from effort incentives, and derives the supply of self-deception from strategic memory under Bayesian reasoning. This bridges psychology and economics without simply assuming irrational beliefs.

10.4 Implications

The framework applies to education, management, sports, entrepreneurship, personal rules, and organizational motivation. Anyone who benefits from another person's success may want to shape that person's self-confidence. Similarly, individuals may design their own information environment to preserve willpower.

10.5 Limitations

The model is theoretical and stylized. It abstracts from many empirical complexities, including social learning, repeated external feedback, heterogeneous tasks, and direct neurological or psychological mechanisms. It also does not estimate the size of memory distortions empirically.

10.6 Takeaway

The lasting lesson is that belief management is part of motivation. Self-confidence can be a productive asset, but managing it can create overconfidence, self-handicapping, and costly self-deception. Rationality and motivated belief need not be opposites; motivated belief can arise from rational strategic interaction between temporal selves.